TAUTOMERISATION DU 4'-DIMETHYLAMINOFLAVONOL A L'ETAT EXCITE: LE TRANSFERT DE PROTON EST-IL UN PROCESSUS REVERSIBLE?

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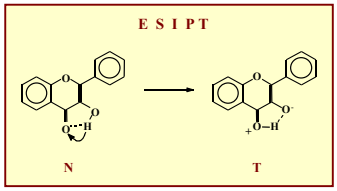
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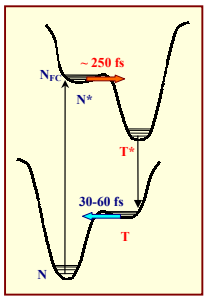
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INTRODUCTION

1

3-hydroxyflavones (flavonols) are very widespread natural compounds, which are useful models for investigation of the excited state proton transfer (ESIPT).





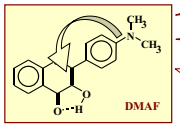
ESIPT in unsubstituted flavonol is:

- very fast process with k ~ from $4.2 \cdot 10^{12}$ (in methylcyclohexane) till $1.7 \cdot 10^{11}$ (acetonitrile)
- irreversible process: only N*→T* in the excited state, only T→N in the ground state
- proceeding by tunnel mechanism in non-polar aprotic solvents (only T band in emission spectra)
- proceeding by tunnel and Arrhenius pathway in polar aprotic solvents (double N/T fluorescence)

2

Excitation of flavonol derivatives – 4'-dialkylaminoflavonols is accompanied by inter-fragmental charge transfer. "Coexistence" charge and proton transfer in the excited state explains researcher's interest to spectral properties of these substances.

Now 4'-dialkylaminoflavonols are used in different spheres of biology, chemistry and physics.



- Radiation hard-scintillator counter
- Laser dye with ASE effect
- Fluorescent probe for analysis of protein conformation, cellular membrane polarity, micelles structure.

used 'analytical signals'

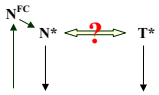
- Shift of N emission band maximum
- Difference between N and T bands maxima
- Ratio of band intensities I_T/I_N

Particularity of spectral behavior of 4'-dialkylaminoflavonols

Double fluorescence in non-polar aprotic solvents : presence of very week emission band of N form

To explain double fluorescence in emission spectra, a hypothesis about **REVERSIBILITY** of excited state proton transfer was proposed

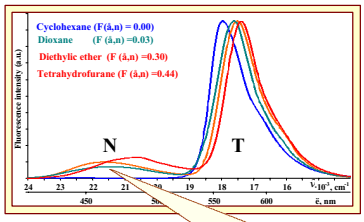
Is it really ESIPT process in DMAF is REVERSIBLE?



Steady-state observations

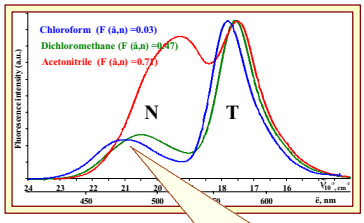
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There is no regular dependency between quantum yields of N and T forms and environment properties.



A

In these solvents the intensity of band N is always low and does not depend on solvent polarity



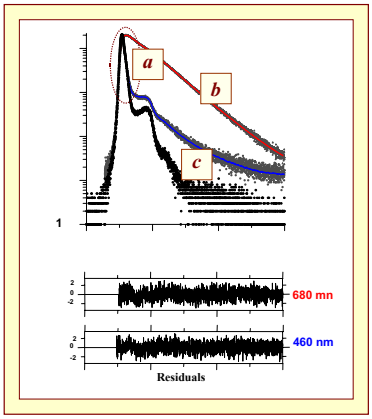
B

In these solvents the intensity of band N depends on environment polarity

Time resolved observations

4

Behavior of excited DMAF in solvents of the group *A*



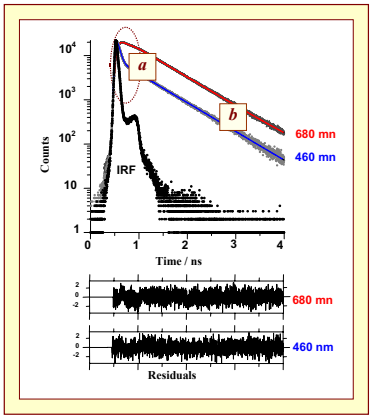
- ***a*** There are a fast decay at 460 nm and corresponding fast rise at 680 nm. That is due to direct ESIPT N→T
- ***b*** There is a slow decay at 680 nm corresponding to quenching of form T.
- ***c*** There is a very week slow decay at 460 nm corresponding to quenching of complexes between N form and solvent molecules.

In solvents of the group *A* ESIPT process is **IRREVERSIBLE** **N*→T***

Time resolved observations

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Behavior of excited DMAF in solvents of the group *B*



- ***a*** There are a fast decay at 460 nm and corresponding fast rise at 680 nm. That is due to direct ESIPT:N→T
- ***b*** There are similar slow decays at 460 and 680 nm. That is due to reverse ESIPT process: T→N

In solvents of the group *B* ESIPT process is **REVERSIBLE** **N*↔T***

Calculation of ESIPT rates

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1. In the case of irreversible ESIPT the rate of the process is equal to value of fast decay/rise component.

2. In the case of reversible ESIPT the rate of the direct and reverse processes were obtained according of Birks' scheme.

From kinetic equations for biexponentials decays obtained at 460 (B) and 680 (R) nm

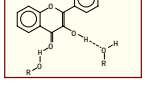
$$I_B = A_{1B}e^{-\frac{t}{\tau_1}} + A_{2B}e^{-\frac{t}{\tau_2}} \quad \text{and} \quad I_R = A_{1R}e^{-\frac{t}{\tau_1}} + A_{2R}e^{-\frac{t}{\tau_2}},$$

equations to ESIPT rate calculations were obtained:

$$X = \frac{\bar{\tau}_1^{-1} - A\cdot\bar{\tau}_2^{-1}}{1-A}, \text{ where } A = \frac{A_{1R}}{A_{2R}} \text{ at condition that } \frac{A_{1R}}{A_{2R}} = -1,$$

$$Y = \bar{\tau}_1^{-1} + \bar{\tau}_2^{-1} - X \quad \text{and} \quad \bar{\tau}_{DPT}\cdot\bar{\tau}_{RPT} = X\cdot Y - \bar{\tau}_1^{-1}\cdot\bar{\tau}_2^{-1}, \text{ where } X = \bar{\tau}_N^{-1} + \bar{\tau}_{DPT}^{-1}, \; Y = \bar{\tau}_T^{-1} + \bar{\tau}_{RPT}^{-1}$$

To obtain the rates of direct (τ_{DPT}) and reverse (τ_{RPT}) processes it is necessary to use a model structure, whose decay of "normal" (τ_N) or "phototautomer" (τ_T) forms may be determined.
As such model, complexes DMAF-water were used.



τ_N of model system

Rates of equilibrium constant

Table. Rates of direct and reverse ESIPT processes

Solvent	τ_N / ns	τ_T / ps	τ_{DPT} / ps	τ_{RPT} / ps	K (τ_{RPT}/τ_{DPT})
Cyclohexane	-	1100	< 10	-	
Dioxane	-	400	15	-	
Diethylic ether	-	360	11	-	
Tetrahydrofurane	-	308	16	-	
Chloroform	1.63	780	86	980	0.09
Dichloromethane	1.93	576	54	362	0.15
Acetonitrile	0.71	358	70	140	0.53

Irreversible ESIPT

Reversible ESIPT

Rates of direct ESIPT processes are slower than that in unsubstituted flavonol

Reverse ESIPT processes possess rates of the same order than rates of phototautomer quenching τ_T

Thermical effects of ESIPT

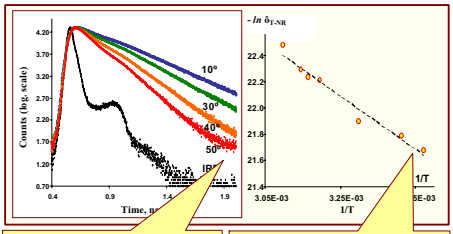
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When DMAF is used as fluorescent probe, the main "analytical signal" is a value of ratio I_N/I_T.

Is it possible to explain I_N/I_T ratio changes only by environment influence on the equilibrium N T in the excited state ?

What is a cause of I_N/I_T ratio changes in the case of irreversible processes?

To answer these questions, thermical effects in the case of irreversible ESIPT were studied (DMAF in THF)



There is no relevant changing direct ESIPT rate under temperature increasing.
On the other hand, the lifetime of T form decreases from 364 to 163 ps

The plot of logarithm of $\hat{o}_{T\text{-}NR}$ value (lifetime due to non-radiative deactivation of T form) vs. 1/T possesses a linear character.
Activation energy is 4,0 kcal/mol

The lack of substantial changing direct ESIPT ratio and, at the same time, a drastic thermical quenching of the T form, which is obtained because of this process, show that the rate of T form quenching is very important factor influencing spectral properties of DMAF and, particularly, I_N/I_T value.

CONCLUSIONS

8

Excited state proton transfer in DMAF and other alkylaminoflavonols may be as REVERSIBLE, as IRREVERSIBLE process depending solvent nature.

Polarity of the solvent is not criterion, which determines the reversibility of proton transfer.

The rate of the direct ESIPT in DMAF (particularly in the case of the reversibility) is substantially lesser than the one in non-substituted flavonol. This effect is due to the fact that phototaumerisation proceeds from charge transferred form N*, where acidity of 3-hydroxy group is significantly lesser.

Constructing new fluorescent probes based on DMAF structure it is important to take into account that parameter I_N/I_T is due not only to equilibrium N↔T in the excited state, but also to phototautomer T form quenching.

ACCNOWGLEMENTS

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